

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We take a look at all we know about Omicron thus far, we cannot figure out why humans have different blood groups, get an update from SpaceX and their attempt to make rocket fuel from CO₂, and look at some fossilised eggs!



World's oldest family tree

Archaeologists from the Universities of Newcastle, Basque Country, Vienna and Harvard have revealed the world's oldest family tree. DNA of 35 individuals was obtained from a 5700-year-old neolithic tomb in Britain. They found five generations of relatives all descended from four women who all had children with the same man.
<https://digit.in/oldfamtree>

WHAT'S NEW

Self-adapting energy-saving glass

Modern cities are full of buildings with glass exteriors, because glass looks good, gives a feeling of openness and doesn't need to be painted and maintained as much. It's terrible at energy-saving though.

An international research team led by scientists from the Nanyang Technological University, Singapore might have developed a solution. It's a new material that is used as a coating and reduces the transfer of heat.

Simulated testing revealed energy savings of up to 9.5% or approximately 3,30,000 kWh a year. The new material can also be customised in terms of its heating and cooling performance according to region and climatic conditions.



The researchers used layers of vanadium dioxide nanoparticles, Poly(methyl methacrylate) (PMMA) and a low-emissivity

coating forming a structure that can modulate heating and cooling simultaneously. In the summertime, the coating can suppress solar heating or near-infrared light, while boosting radiative cooling or long-wave, and the opposite in winter. In short, using spectrums of light to heat or cool the interior.

<https://digit.in/coolglass>

James Webb is in space!

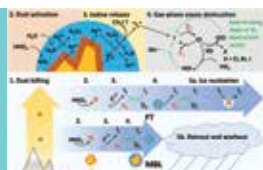
After years of being stuck in the frustrating development stage, NASA's James Webb Telescope, has finally found its way into space. On December 25 2021, the telescope was launched, aboard the Ariane 5 ECS rocket, and now, is on to working towards its goal of revealing

the hidden realities of the formation and development of our universe.



The James Webb is a major step forward in the field of astronomy. Equipped with a primary mirror, which is 6.5 meters (across) in size, and four science instruments, namely Near-Infrared Camera (NIRCam), Near-

Infrared Spectrograph (NIR-Spec), Mid-Infrared Instrument (MIRI), and Near-Infrared Imager and Slitless Spectrograph (NIRISS) with the Fine Guidance Sensor (FGS), this telescope is all set to provide astronomers across the world, some really interesting information about the early phase in the history of the universe, revealing the various secrets of the bang that started it all.
<https://digit.in/jwisup>



Iodine's killing it

A study, led by researchers from the University of Colorado at Boulder, has revealed that iodine dust rising from the deserts into the atmosphere, is playing a significant role in effectively destroying the ozone layer!
<https://digit.in/badiodine>



Planets have gone rogue!

A team of scientists analysed data collected from various telescopes of the European Southern Observatory and successfully identified a group of 70 rogue planets – the largest ever found, in our galaxy.
<https://digit.in/badplanet>



Cancer research can't be replicated

Two research papers published recently, have suggested that cancer research cannot be replicated in any way, commenting on the credibility, maintenance, and use of pre-existing research.
<https://digit.in/elife1> + <https://digit.in/elife2>